SALESFORCE CRM PROJECT DOCUMENTATION

Replastix Innovations

Plastic Waste Recycling and Inventory Management

Project Focus

Salesforce Data Modelling, Salesforce Data Security, Objects, Fields and Relationships, Flow Builder, and Approving Records using Apex.

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1 Project Overview

Replastix Innovations is an organization focused on plastic waste collection, recycling, recycled-product management, order processing, and stock replenishment. The project uses Salesforce to organize these business processes through a structured data model, controlled data access, validation, automation, and Apex-based processing.

The Salesforce implementation represents plastic waste, recycling centers, recycled products, orders, and restock requests as Salesforce objects. The system also controls access to information using roles, profiles, users, and record-level sharing. Automation is used to identify low stock and create a task for the appropriate record owner. Apex is used to process product stock, create restock requests when an order cannot be fulfilled from available stock, and update stock after a restock request is approved.

Business Need

The main business need is to maintain accurate information about recyclable materials and recycled products while reducing manual work in stock management. The Salesforce implementation connects related records and automates important actions when stock becomes insufficient.

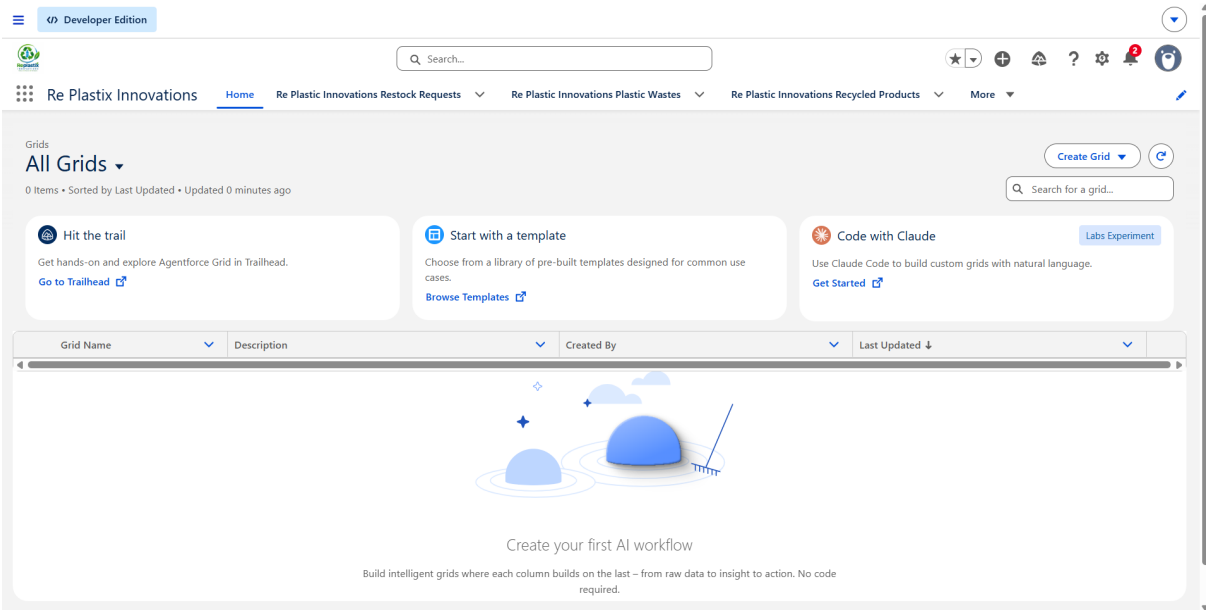


Figure 1: Salesforce environment showing the Replastix Innovations application.

2 Objectives

The objectives of the Replastix Innovations Salesforce project are:

- To create a structured Salesforce data model for plastic waste, recycling centers, recycled products, orders, and restock requests.
- To establish relationships between the objects so that related business information can be accessed efficiently.
- To protect business data by assigning appropriate roles, profiles, users, and record-level access.
- To validate important business data, such as order quantity and collection date.
- To automate low-stock identification and task creation using Flow Builder.
- To use Apex to process orders, create restock requests when stock is insufficient, and update product stock after approval.
- To support accurate and consistent inventory management through Salesforce automation and development features.

3 Phase 1: Requirement Analysis and Planning

3.1 Understanding Business Requirements

The project is designed around the operational requirements of a plastic recycling organization. The system needs to maintain information about plastic waste and recycling centers, track recycled products and their stock, manage customer orders, and handle replenishment requests when stock is insufficient.

The project also requires controlled access to records. Different users perform different responsibilities, so Salesforce security is configured to provide different levels of access.

3.2 Project Scope

The scope of the implementation is limited to the project-description topics:

- Salesforce data modelling.
- Salesforce data security.
- Objects, fields, and relationships.
- Flow Builder and process automation.

- Approval-related stock processing using Apex.

Topics such as reports and dashboards, Lightning Web Components, data migration, deployment strategy, and other features mentioned only in the general documentation guidelines are intentionally not included because they are not part of the stated project-description learning topics.

3.3 Data Model Design

The main custom objects created for the project are:

1. Re Plastic Innovations Plastic Waste
2. Re Plastic Innovations Recycling Center
3. Re Plastic Innovations Recycled Product
4. Re Plastic Innovations Order
5. Re Plastic Innovations Restock Request

The standard **Account** object is also used as the customer object for orders.

3.4 Security Model Design

The security model uses:

- Roles and role hierarchy.
- Profiles.
- Users.
- Record-level security and sharing rules.

The configured hierarchy places the CEO above the Recycling Manager and Sales Representative, with the Warehouse Supervisor below the Sales Representative, as represented in the project role hierarchy.

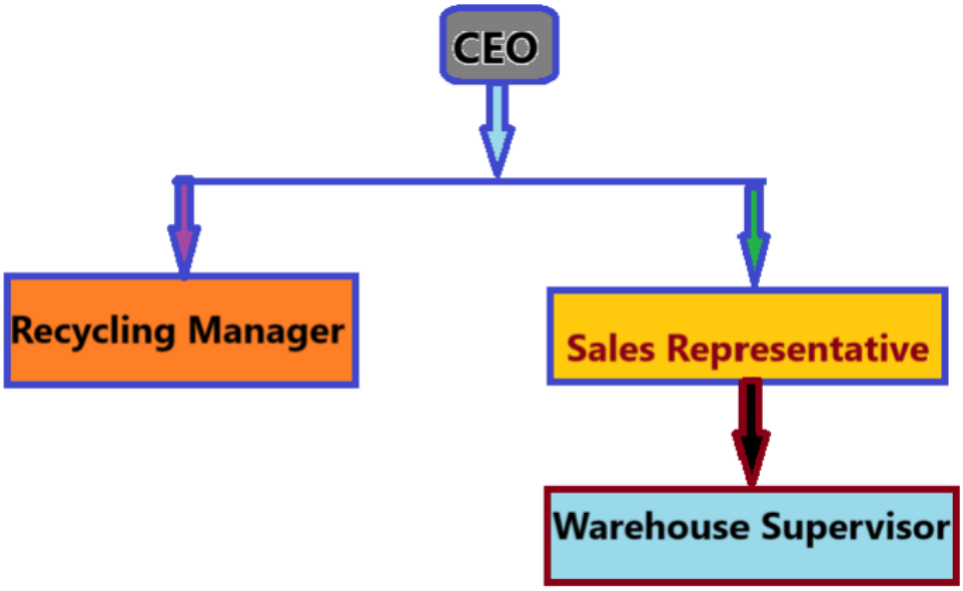


Figure 2: Configured Salesforce role hierarchy for Replastix Innovations.

4 Salesforce Data Modelling

4.1 Custom Objects

The following custom objects represent the major entities in the system.

Object	Purpose
Re Plastic Innovations Plastic Waste	Stores information about collected plastic waste, including weight, plastic type, collection date, status, recycling center, and location.
Re Plastic Innovations Recycling Center	Stores recycling-center information such as center name, location, and processing capacity.
Re Plastic Innovations Recycled Product	Stores recycled products and their inventory information, including stock level, threshold, and price.
Re Plastic Innovations Order	Stores customer orders, ordered products, quantities, and delivery dates.
Re Plastic Innovations Restock Request	Stores replenishment requests for recycled products, requested quantity, and approval status.

Re Plastic Innovations Order	Re_Plastic_Innovations_Order__c	Custom Object	Order Details	9/19/2026	✓	▾
Re Plastic Innovations Plastic Waste	Re_Plastic_Innovations_Plastic_Waste__c	Custom Object	Unique ID for waste records	9/16/2026	✓	▾
Re Plastic Innovations Recycled Product	Re_Plastic_Innovations_Recycled_Product__c	Custom Object	Recycled product name	9/16/2026	✓	▾
Re Plastic Innovations Recycling Center	Re_Plastic_Innovations_Recycling_Center__c	Custom Object	Recycling Center Information	9/19/2026	✓	▾
Re Plastic Innovations Restock Request	Re_Plastic_Innovations_Restock_Request__c	Custom Object	Request ID	9/16/2026	✓	▾

Figure 3: Custom Objects created for the Replastix Innovations data model.

4.2 Fields: Plastic Waste

Object: **Re Plastic Innovations Plastic Waste**

Field API Name	Data Type	Description
Name	Text (Auto Number)	Unique ID for waste records.
`Weight__c`	Number (18,2)	Weight of plastic waste in kilograms.
`Type__c`	Picklist	Type of plastic, such as PET, HDPE, PVC, etc.
`Collection_Date__c`	Date	Date on which the waste was collected.
`Status__c`	Picklist	Collected, Processing, or Recycled.
`Recycling_Center__c`	Lookup	Assigned recycling center.
`Location__c`	Geolocation	Waste collection location.

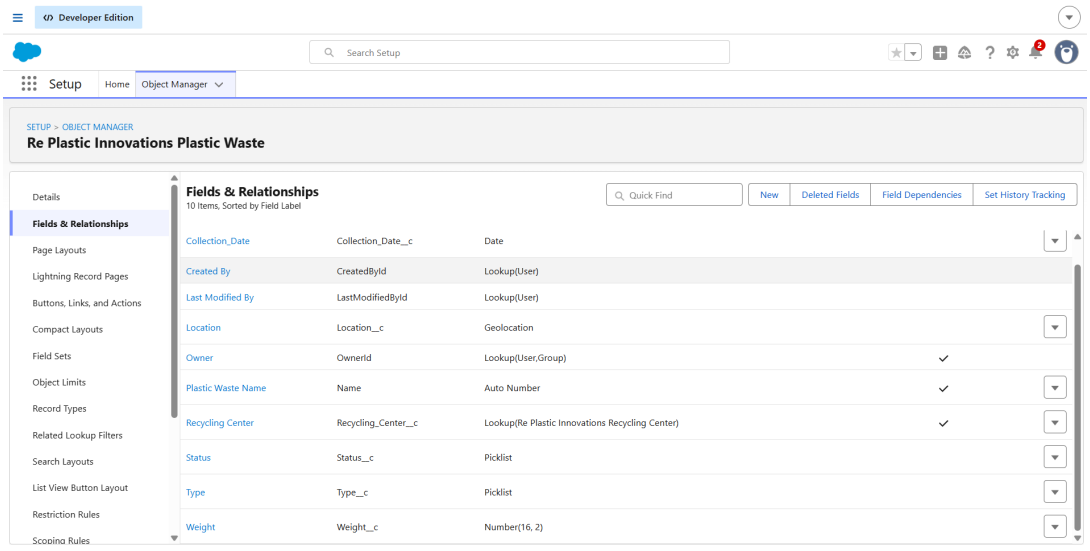


Figure 4: Fields and relationships configured for the Plastic Waste object.

4.3 Fields: Recycling Center

Object: **Re Plastic Innovations Recycling Center**

Field API Name	Data Type	Description
Name	Text	Recycling center name.
`Location__c`	Geolocation	Location of the recycling center.
`Capacity__c`	Number (18,2)	Maximum processing capacity.

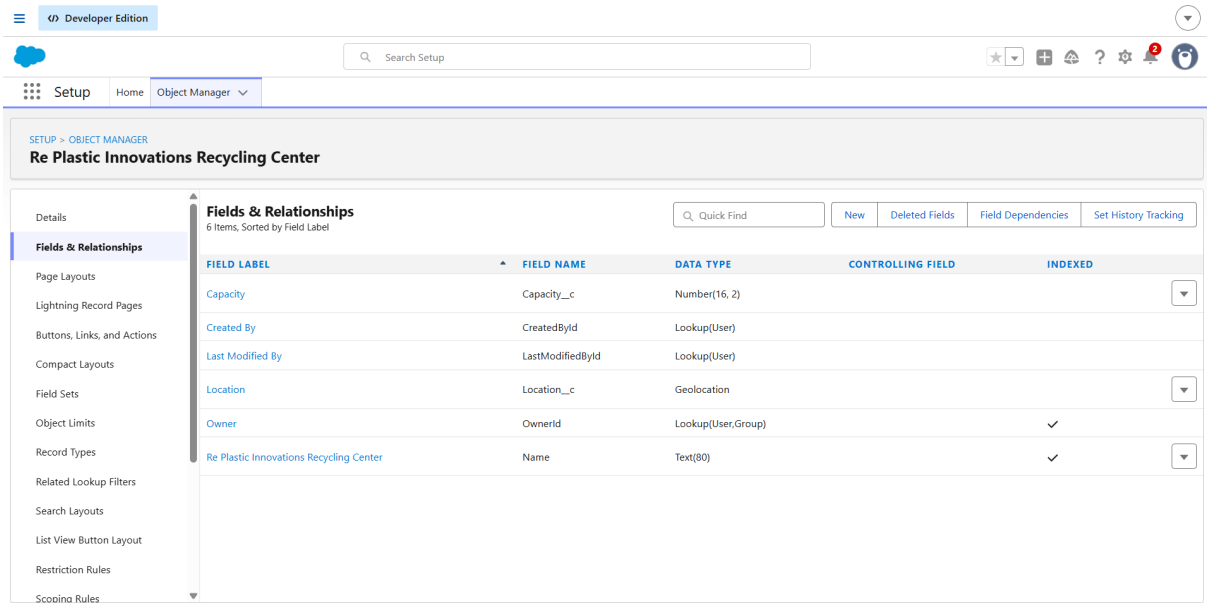


Figure 5: Fields configured for the Recycling Center object.

4.4 Fields: Recycled Product

Object: **Re Plastic Innovations Recycled Product**

Field API Name	Data Type	Description
Name	Text	Recycled product name.
`Stock_Level__c`	Number	Current stock available.
`Threshold__c`	Number	Minimum stock before restock is required.
`Price__c`	Currency	Price per unit.

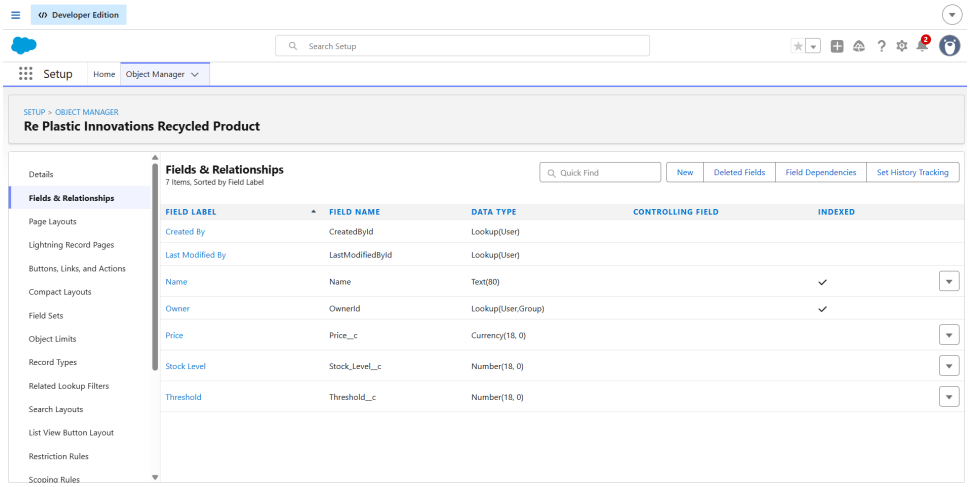


Figure 6: Fields configured for the Recycled Product object.

4.5 Fields: Order

Object: **Re Plastic Innovations Order**

Field API Name	Data Type	Description
Name	Auto Number	Order ID.
`Customer__c`	Lookup (Account)	Customer placing the order.
`Recycled_Product__c`	Lookup	Ordered recycled product.
`Quantity__c`	Number	Quantity ordered.
`Delivery_Date__c`	Date	Expected delivery date.

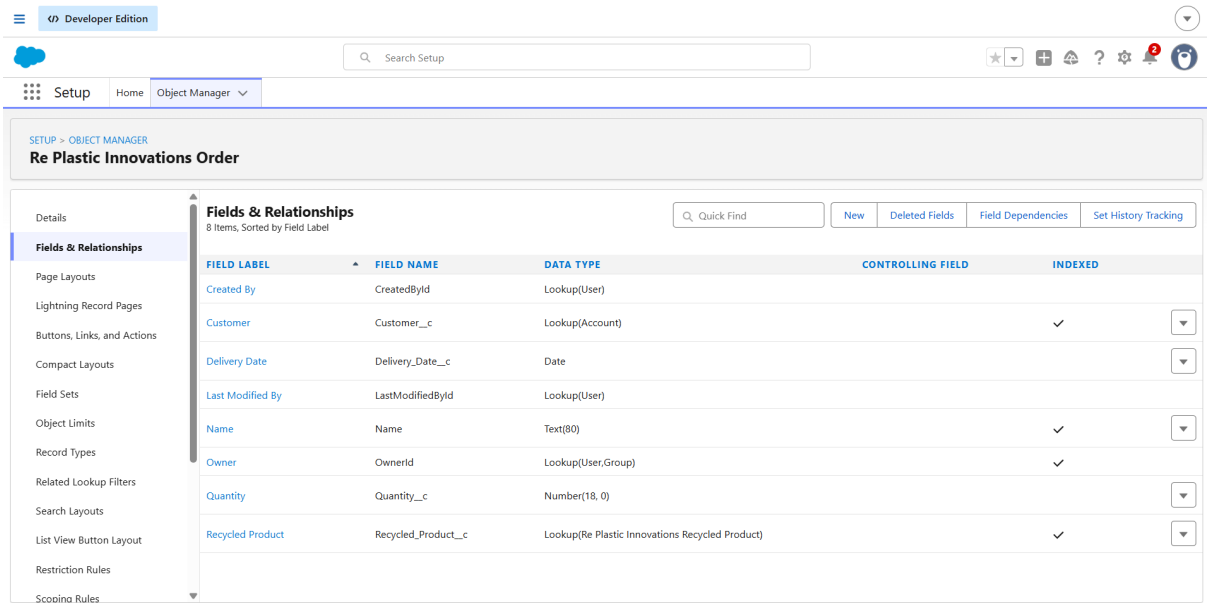


Figure 7: Fields configured for the Order object.

4.6 Fields: Restock Request

Object: **Re Plastic Innovations Restock Request**

Field API Name	Data Type	Description
Name	Auto Number	Restock request ID.
`Product__c`	Lookup	Product that requires replenishment.
`Requested_Quantity__c`	Number	Quantity requested for replenishment.
`Status__c`	Picklist	Pending, Approved, or Completed.

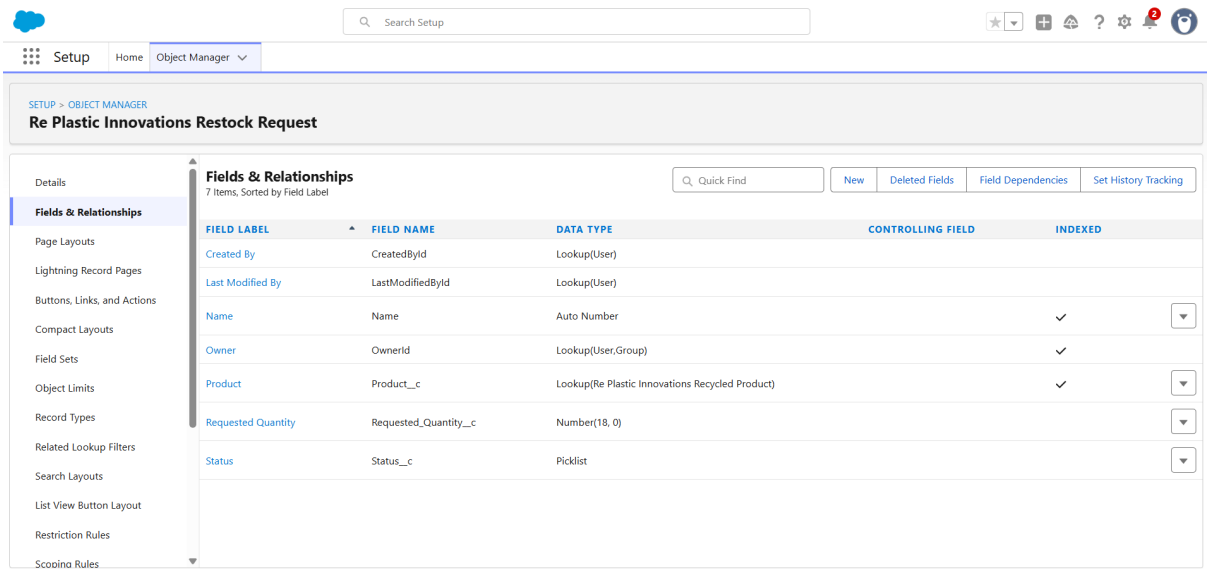


Figure 8: Fields configured for the Restock Request object.

4.7 Object Relationships

The important relationships in the data model are:

- **Plastic Waste → Recycling Center:** Plastic Waste contains a lookup field to the Recycling Center object.
- **Order → Account:** Each order can reference the customer through the Account object.
- **Order → Recycled Product:** Each order references the recycled product being ordered.
- **Restock Request → Recycled Product:** Each restock request references the product that needs replenishment.

These relationships connect the operational records and allow the product record to be associated with orders and restock requests.

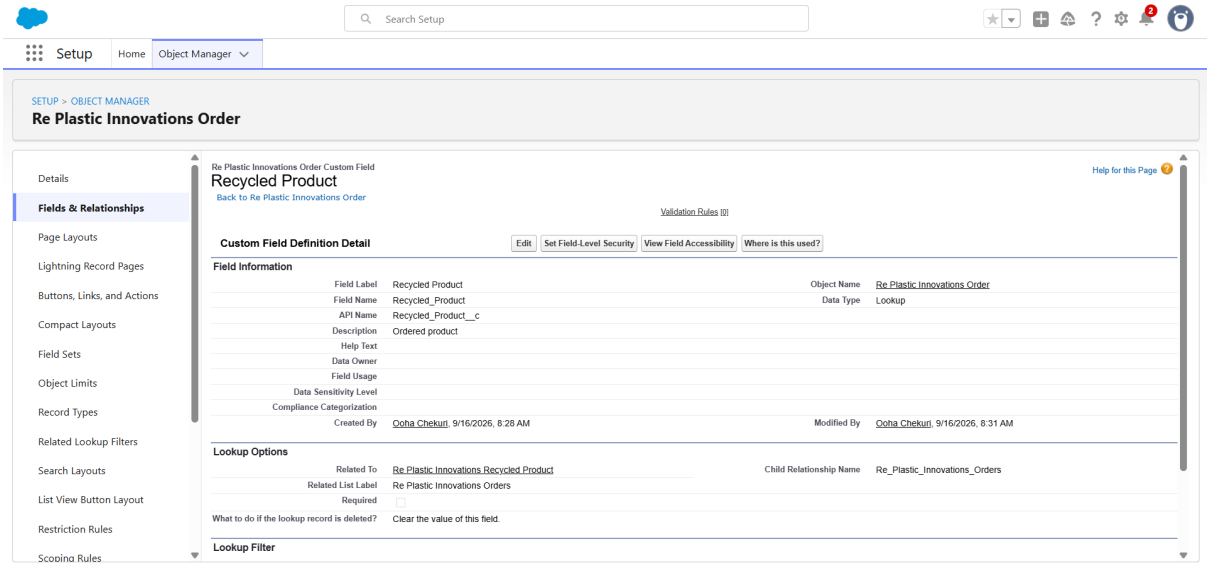


Figure 9: Relationship between Order and Recycled product

5 Salesforce Data Security

5.1 Roles and Role Hierarchy

The configured role hierarchy contains the following roles:

- CEO
- Recycling Manager
- Sales Representative
- Warehouse Supervisor

The hierarchy is used as part of the organization's record-level access model.

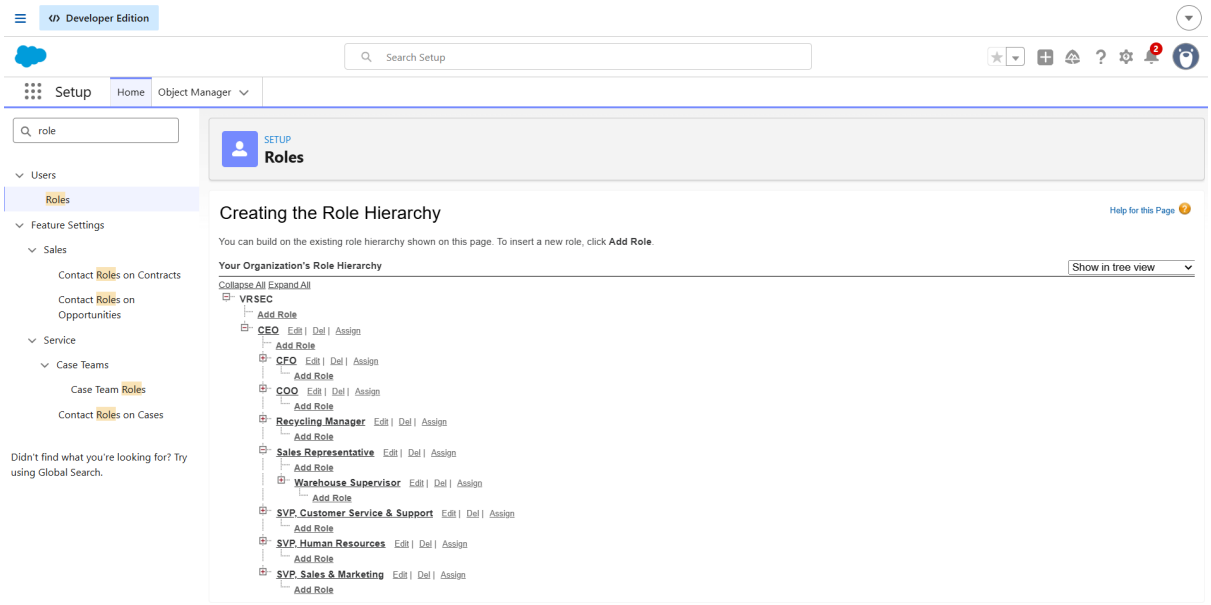


Figure 10: Salesforce Roles and Role Hierarchy configuration

5.2 Profiles

Three profiles were created and configured according to the required access levels.

5.2.1 Platform 1

- Re Plastic Innovations Plastic Waste: Read/Create.
- Re Plastic Innovations Restock Request: Read-Only.

5.2.2 Platform 2

- Re Plastic Innovations Order: Read/Create.
- Account: Read/Create.
- Re Plastic Innovations Plastic Waste: Read-Only.
- Re Plastic Innovations Recycled Product: Read-Only.

5.2.3 Platform 3

- All custom objects: Read/Create/Edit.

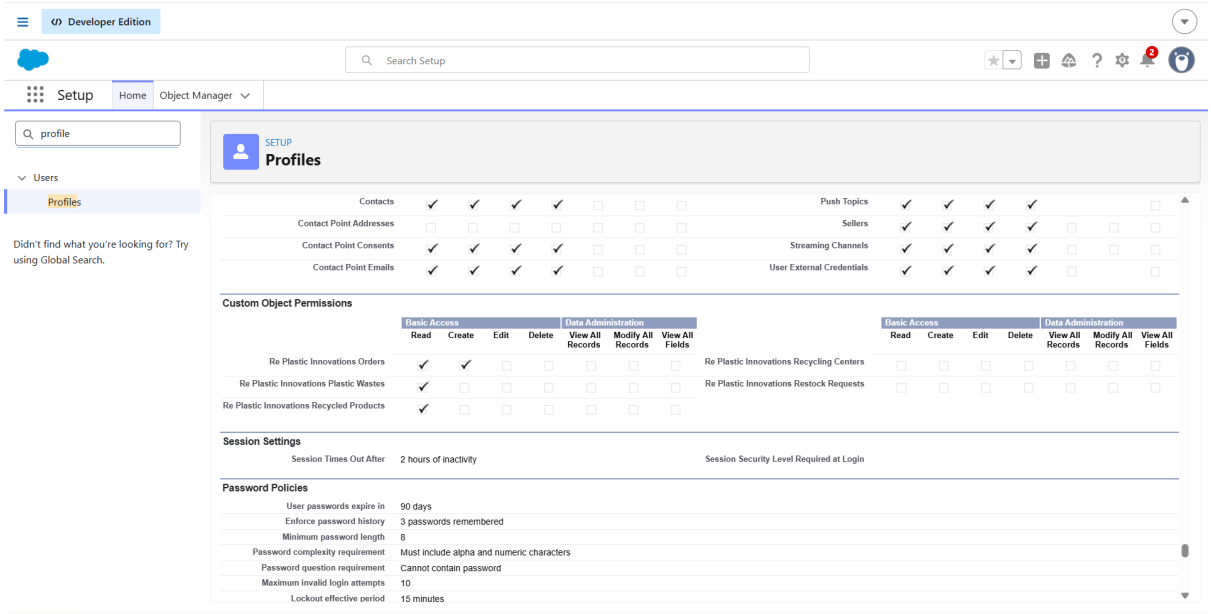


Figure 11: Platform 2 profile configuration

5.3 Users

Three users were configured with the following role and profile assignments.

User	Role	Profile	License
John Production Engineer Sandbox 1	Recycling Manager	Platform 1	Salesforce Platform
Quality Inspector Mike	Sales Representative	Platform 2	–
Plant Manager Albert	Warehouse Supervisor	Platform 3	–

User email addresses are configured using the required personal email addresses, and usernames follow the Salesforce username format specified for the project.

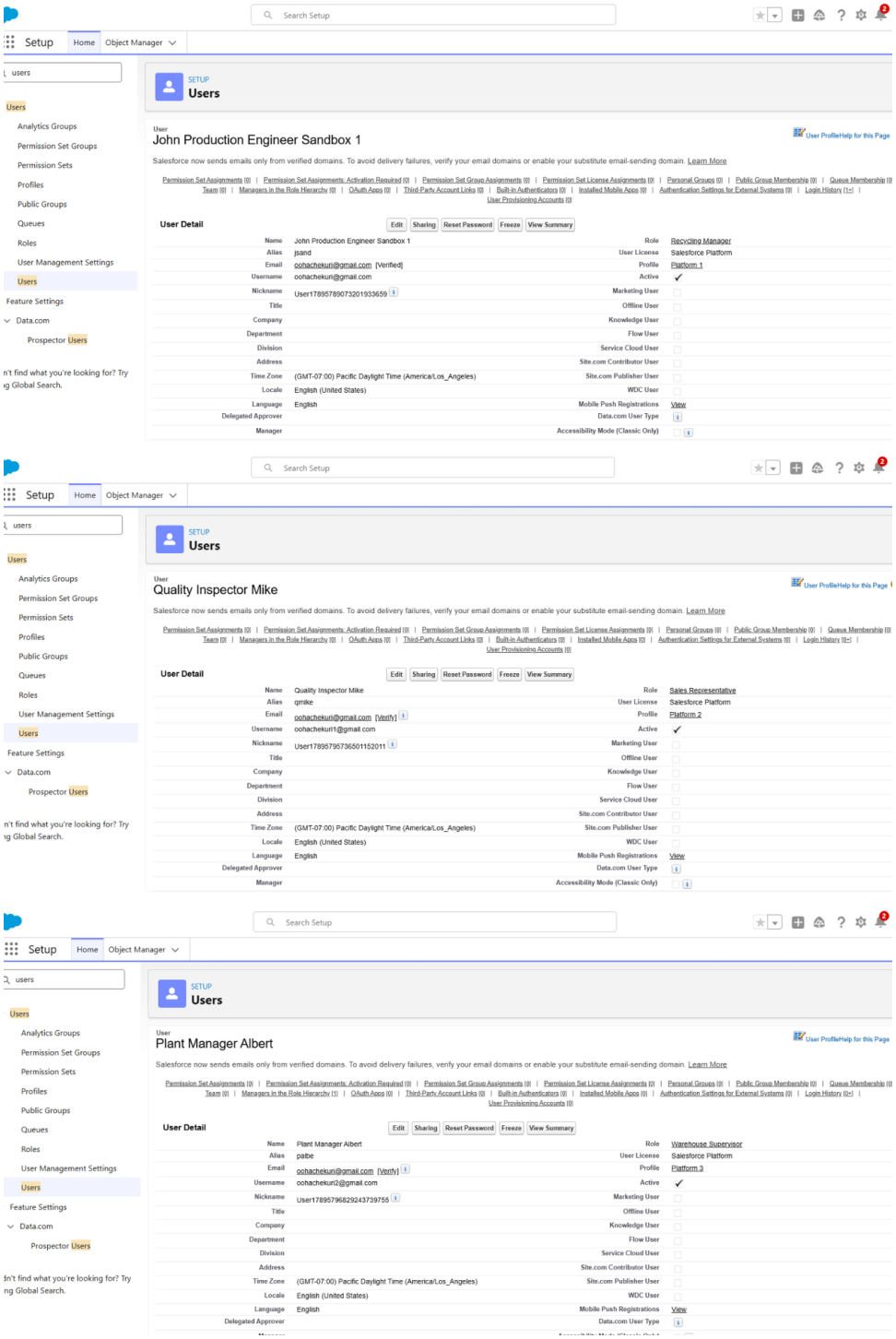


Figure 12: Configured Salesforce users, roles, and profiles.

5.4 Record-Level Security

The following record-level sharing configurations were created.

1. **Recycling Manager Access**

   Label: Recycling Manager Access.

Rule Name: `Recycling_Manager_Access`.

The configured sharing rule provides Read Only access to the Recycling Manager for the selected CEO records.

2. **Sales Representative Access**

   The Sales Representative receives Read access to the Re Plastic Innovations Recycled Product object records as configured.

3. **Warehouse Supervisor Access**

   The Warehouse Supervisor receives Read access to Re Plastic Innovations Restock Request records through the configured sharing relationship.

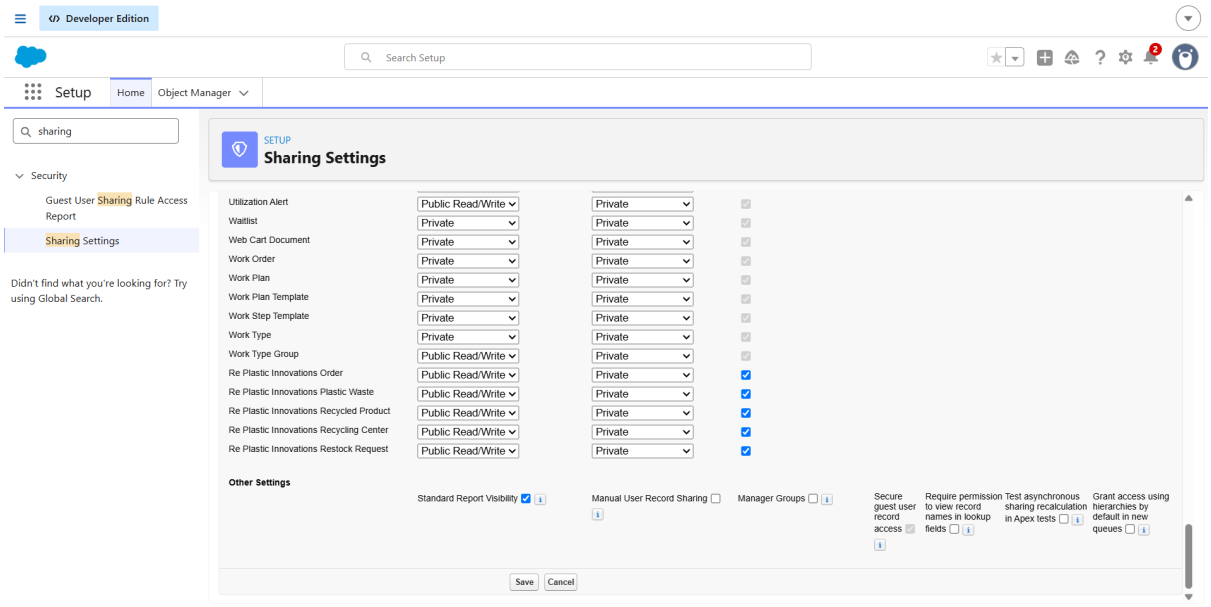


Figure 13: Record-level sharing configuration for the required users and records.

6 Flow Builder

6.1 Purpose of Automation

Flow Builder is used to automate inventory monitoring. The main automation checks whether the current stock of a recycled product is below its defined threshold. When the condition is satisfied, Salesforce creates a Task associated with the product and assigns the task to the product record owner.

6.2 Formula Field: Stock Low On Product

A formula field was created on the Re Plastic Innovations Recycled Product object.

- **Label:** Stock Low On Product
- **Data Type:** Text

The formula used is:

```
IF(Stock_Level__c < Threshold__c,
   "Low Stock - Restock Needed",
   "Sufficient Stock")
```

The formula displays a low-stock message when the current stock is below the configured threshold; otherwise, it displays that the stock is sufficient.

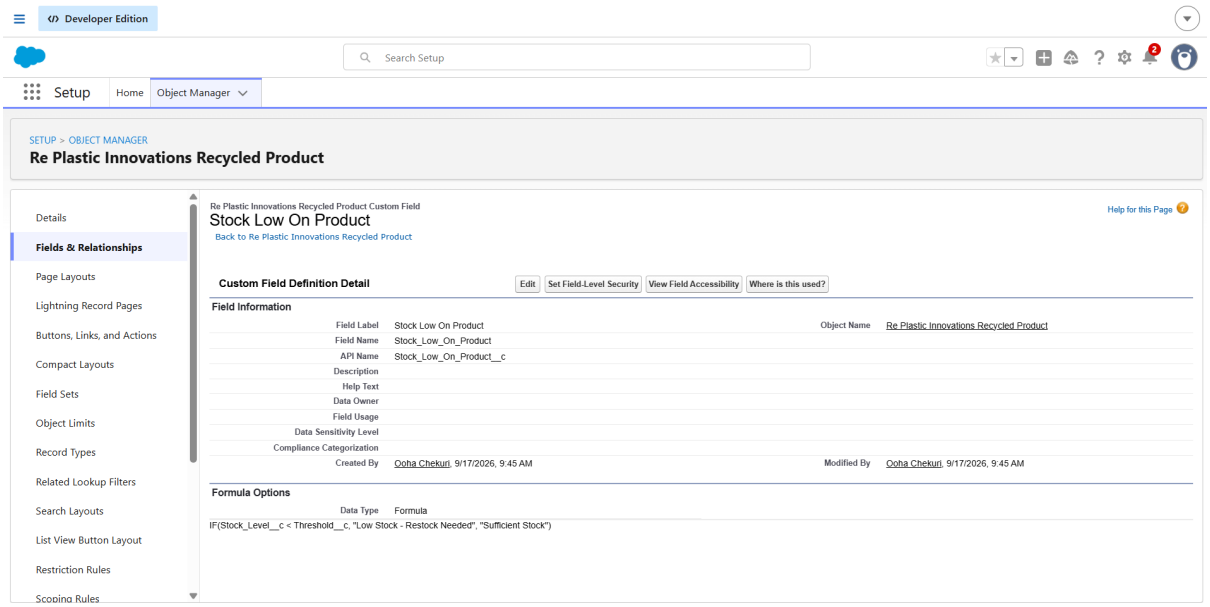


Figure 14: Formula field used to identify whether a product has low stock.

6.3 Validation Rule: Order Quantity

A validation rule was created on the Re Plastic Innovations Order object.

- **Rule Name:** `Check_Quantity_Not_Zero`
- **Active:** True
- **Formula:** `Quantity__c <= 0`

- **Error Message:** Quantity must be greater than zero.
- **Error Field:** Quantity

The rule prevents an order from being saved when the quantity is zero or negative.

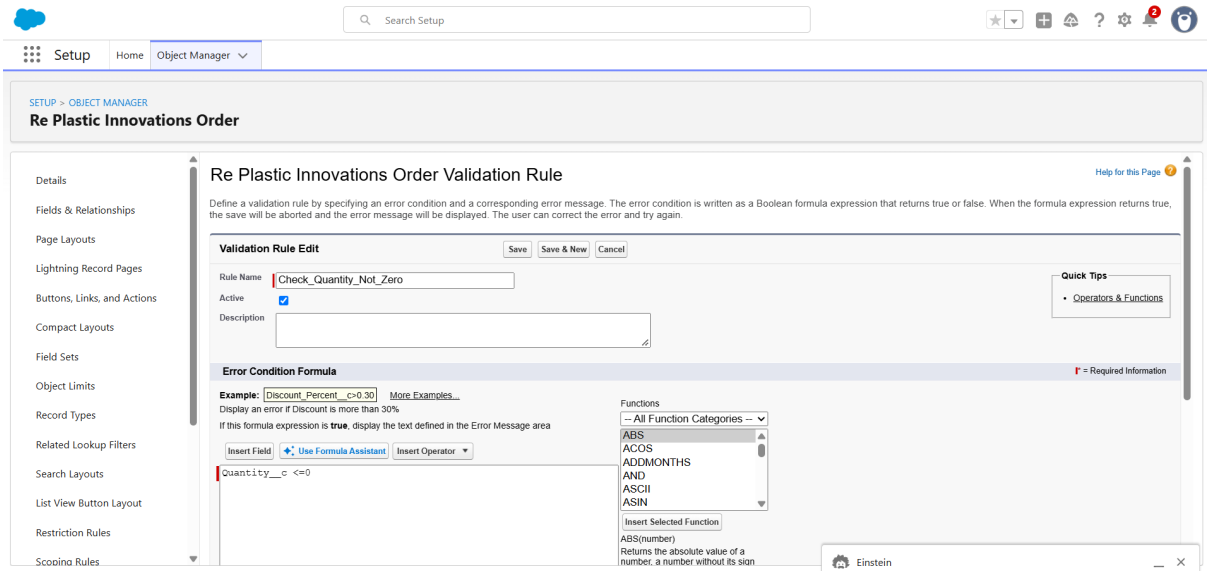


Figure 15: Validation rule preventing zero or negative order quantities.

6.4 Validation Rule: Future Collection Date

A second validation rule was created on the Re Plastic Innovations Plastic Waste object.

- **Rule Name:** `Future_Date_Collection`
- **Active:** True
- **Formula:** `Collection_Date__c > TODAY()`
- **Error Message:** Collection Date cannot be in the future.
- **Error Field:** Collection Date

The rule prevents users from entering a future date as the collection date.

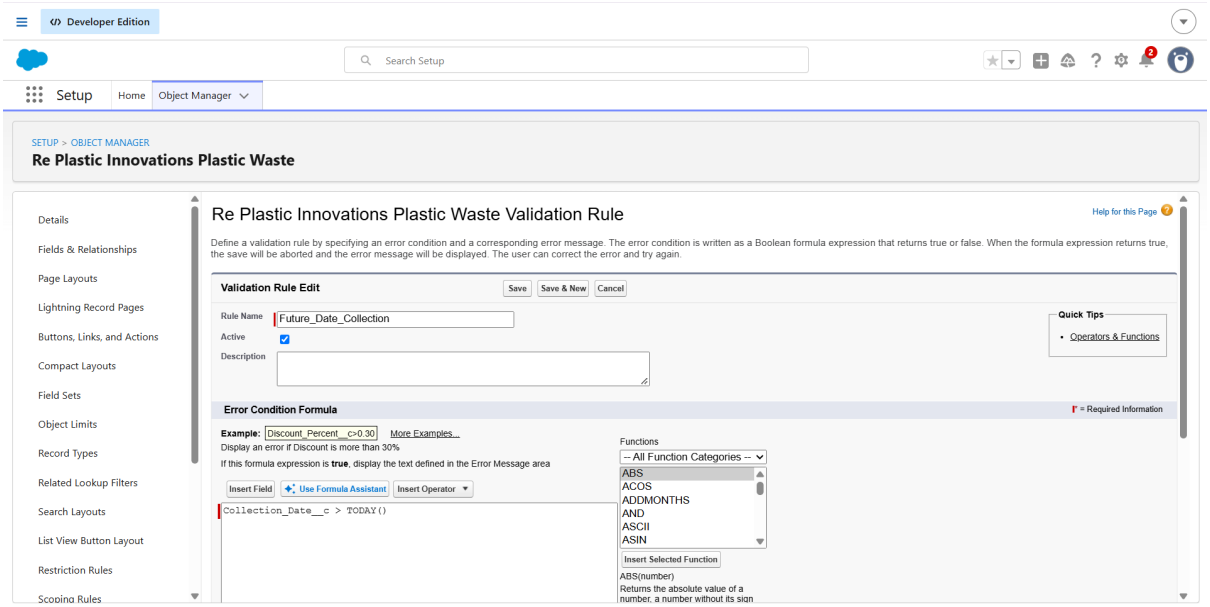


Figure 16: Validation rule preventing a future collection date

6.5 Scheduled Flow: Stock Level Is Low

A scheduled Flow was created for the Re Plastic Innovations Recycled Product object.

Configuration	Value
Flow Type	Schedule-Triggered Flow
Object	Re Plastic Innovations Recycled Product
Frequency	Daily
Time	6:00 AM
Decision Condition	Stock Level is less than Threshold
Outcome	Outcome 1 of Decision 1
Task Object	Task
Task Owner	Product Record Owner
Task Priority	High
Task Status	In Progress
Task Subject	Please Look in this Stock Is Low Fill The Stock ASAP
Flow Label	Stock Level Is Low

The decision checks the following condition:

```
$Record.Stock_Level__c < $Record.Threshold__c
```

When the condition is true, a Task record is created with:

- **Assigned To ID:** `\$Record.OwnerId`
- **Related To ID:** `\$Record.Id`
- **Priority:** High
- **Status:** In Progress
- **Subject:** Please Look in this Stock Is Low Fill The Stock ASAP

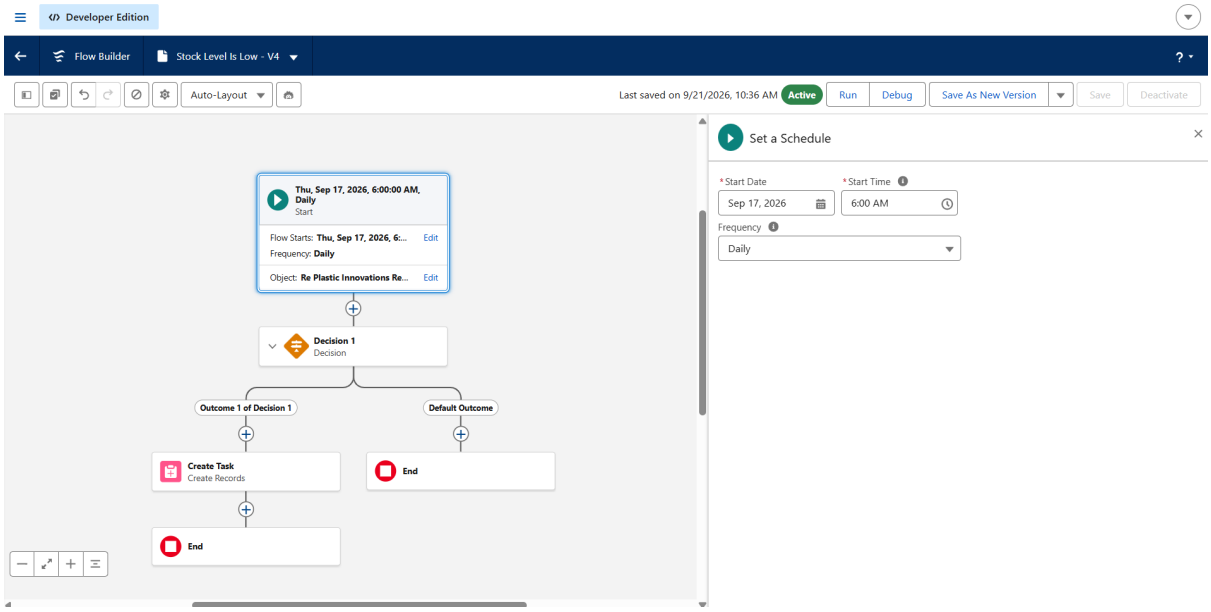


Figure 17: Scheduled Flow configuration and start settings.

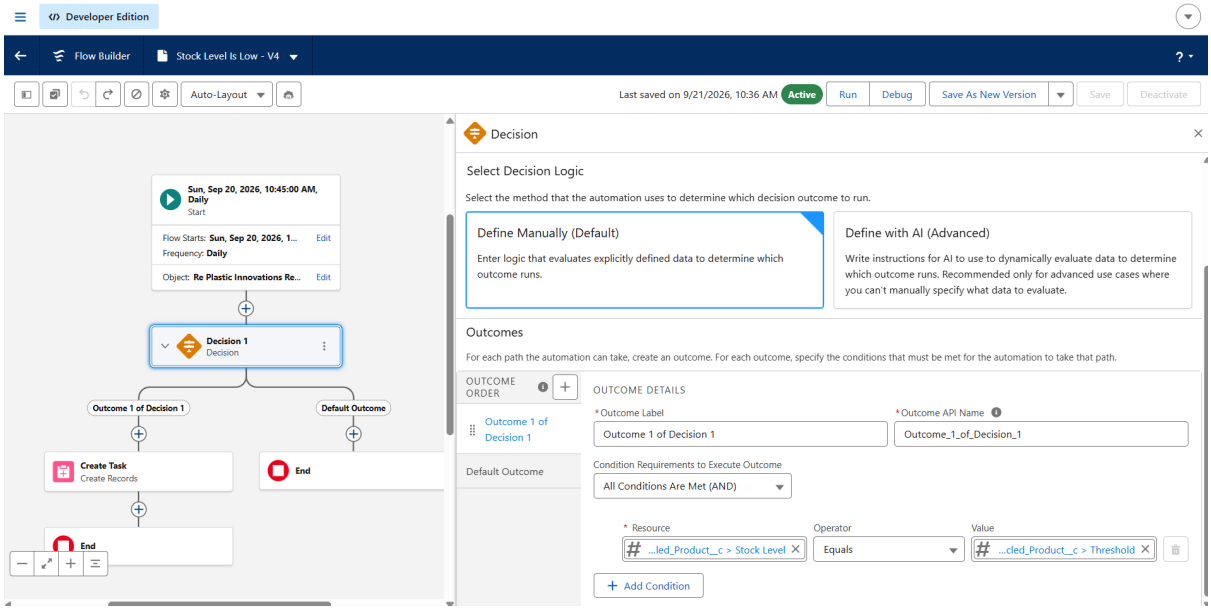


Figure 18: Decision element checking stock level against the threshold

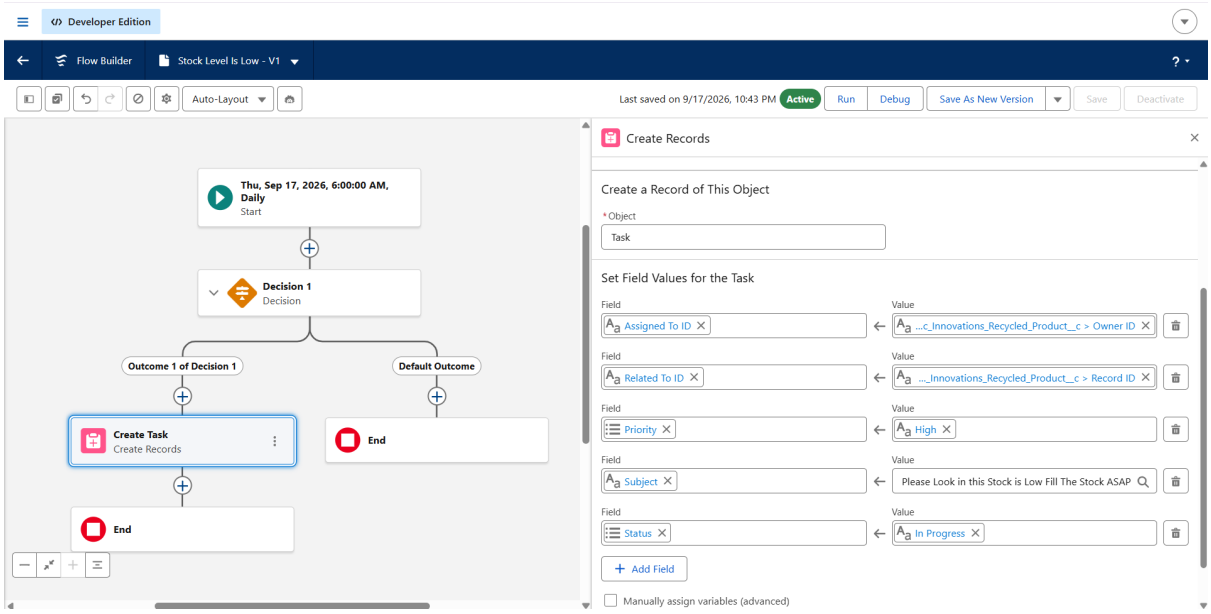


Figure 19: Create Records element used to create the low-stock Task

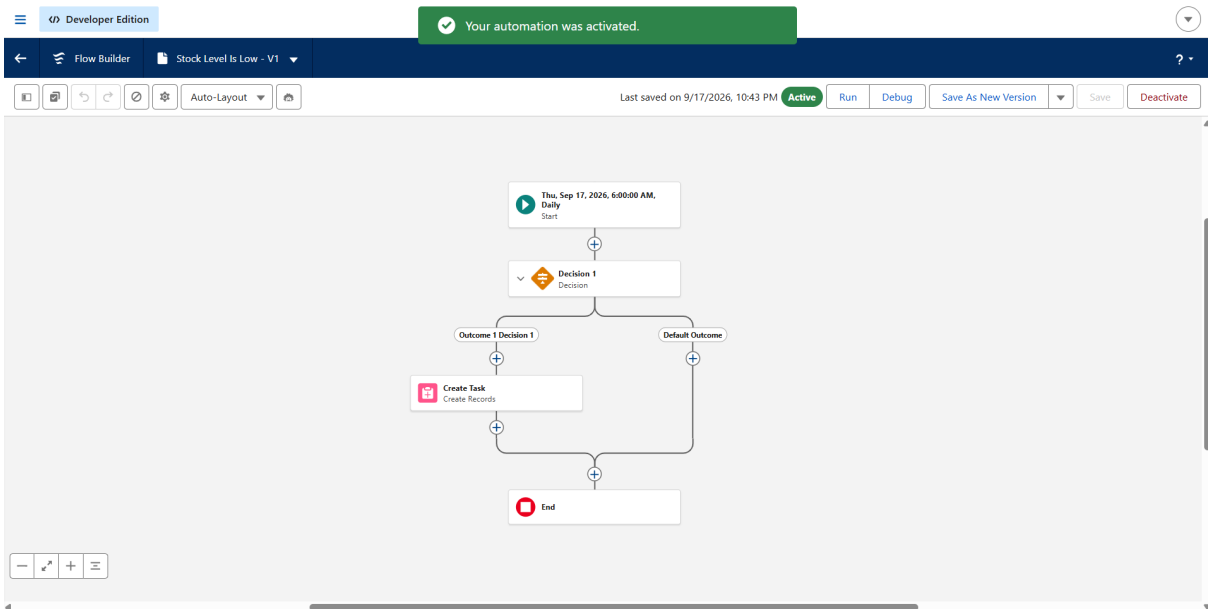


Figure 20: Complete Stock Level Is Low Flow

7 Approving Records Using Apex

7.1 Apex Inventory Processing

The project uses Apex to automate inventory processing when orders are created and when restock requests are approved.

The main Apex class is:

InventoryManager

It contains two main processing methods:

1. `processOrderStock`
2. `processRestockApproval`

7.2 processOrderStock

The `processOrderStock` method processes the stock associated with newly created orders.

Its logic is:

1. Collect the recycled-product IDs referenced by the orders.
2. Query the corresponding recycled-product records.
3. Compare the available stock with the ordered quantity.
4. If sufficient stock is available, reduce the product stock by the ordered quantity.
5. If the available stock is insufficient, create a Restock Request.
6. Set the Restock Request status to Pending.
7. Store the quantity required for replenishment.

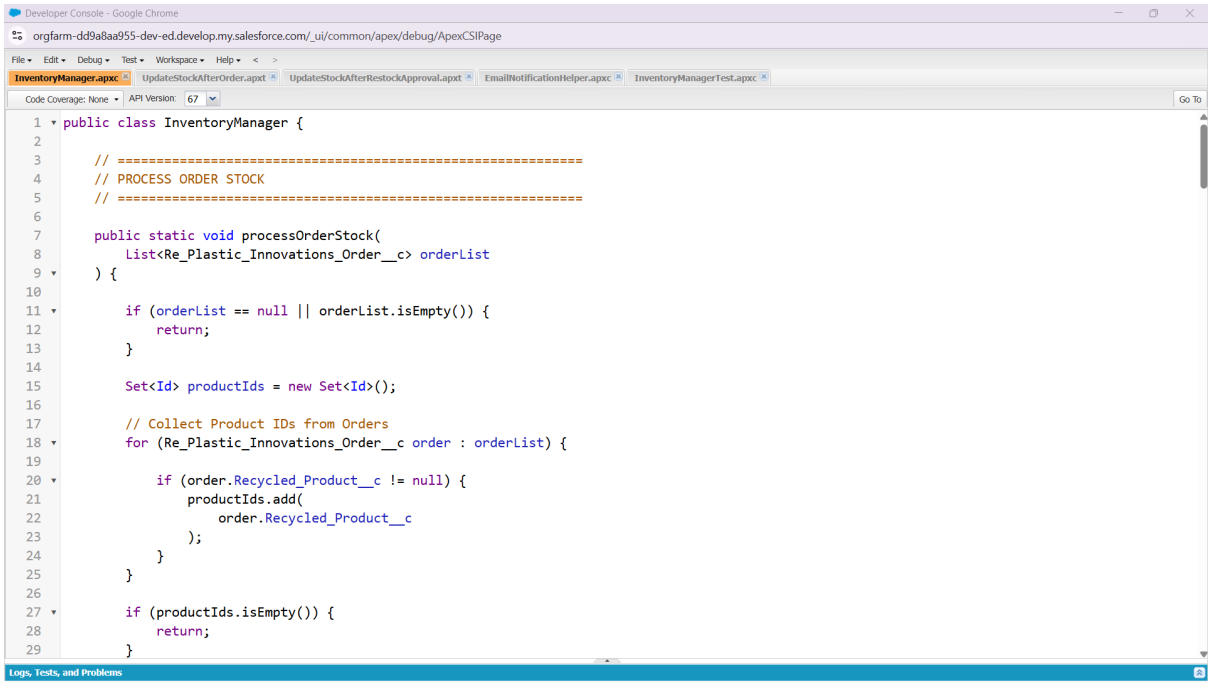


Figure 21: InventoryManager Apex class.

7.3 `UpdateStockAfterOrder` Trigger

The `UpdateStockAfterOrder` trigger runs after an order is inserted.

Its purpose is to call the inventory-processing method:

```
trigger UpdateStockAfterOrder
on Re_Plastic_Innovations_Order__c (after insert) {

    if (Trigger.isAfter && Trigger.isInsert) {
        InventoryManager.processOrderStock(Trigger.new);
    }
}
```

This connects order creation with the inventory-processing logic.

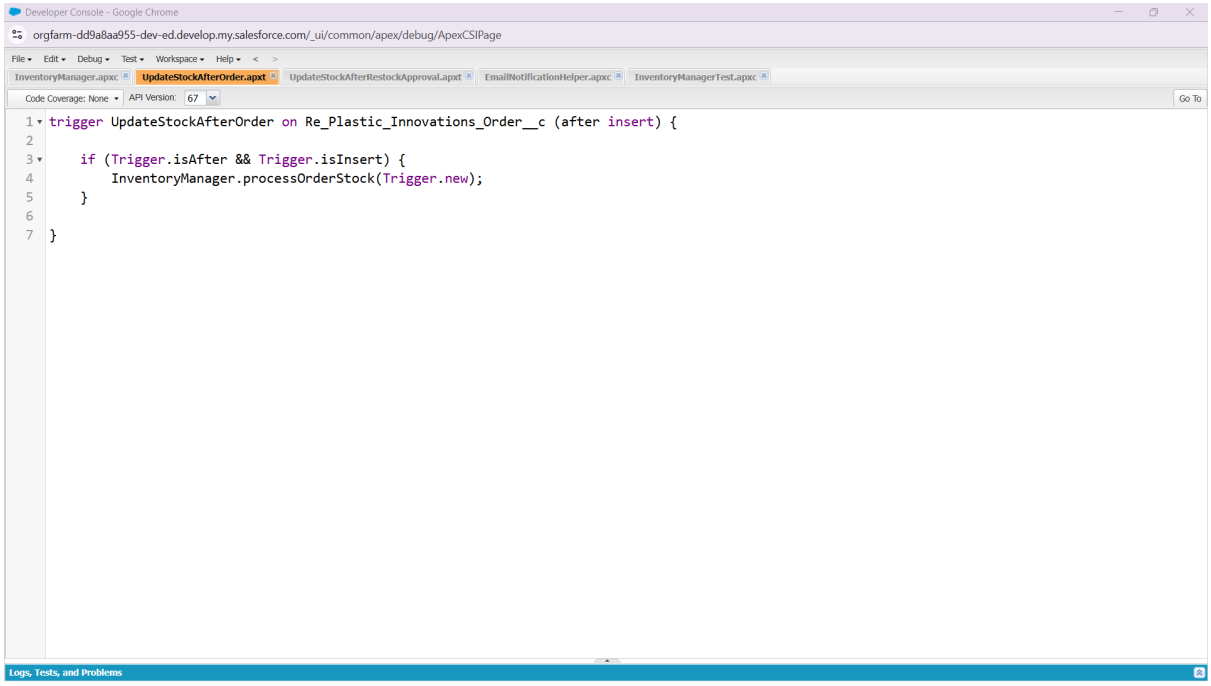


Figure 22: UpdateStockAfterOrder Apex trigger.

7.4 `processRestockApproval`

The `processRestockApproval` method processes approved restock requests.

Its logic is:

1. Identify restock requests whose status is Approved.
2. Collect the related recycled-product IDs.

3. Query the corresponding recycled-product records.

4. Add the requested restock quantity to the current product stock.

5. Update the product records.

This allows approved replenishment requests to increase the stock of the related recycled product.

7.5 `UpdateStockAfterRestockApproval` Trigger

The `UpdateStockAfterRestockApproval` trigger runs after a Restock Request is updated.

It identifies records whose status has changed to Approved and passes those records to `processRestockApp`

```
trigger UpdateStockAfterRestockApproval
on Re_Plastic_Innovations_Restock_Request__c (after update) {

    List<Re_Plastic_Innovations_Restock_Request__c> approvedRestocks =
        new List<Re_Plastic_Innovations_Restock_Request__c>();

    for (Re_Plastic_Innovations_Restock_Request__c restock : Trigger.new) {

        if (restock.Status__c == 'Approved' &&
            Trigger.oldMap.get(restock.Id).Status__c != 'Approved') {

            approvedRestocks.add(restock);
        }
    }

    if (!approvedRestocks.isEmpty()) {
        InventoryManager.processRestockApproval(approvedRestocks);
    }
}
```

7.6 Restock Approval Process

The complete Apex-supported approval sequence is:

1. A recycled product has a defined stock level and threshold.

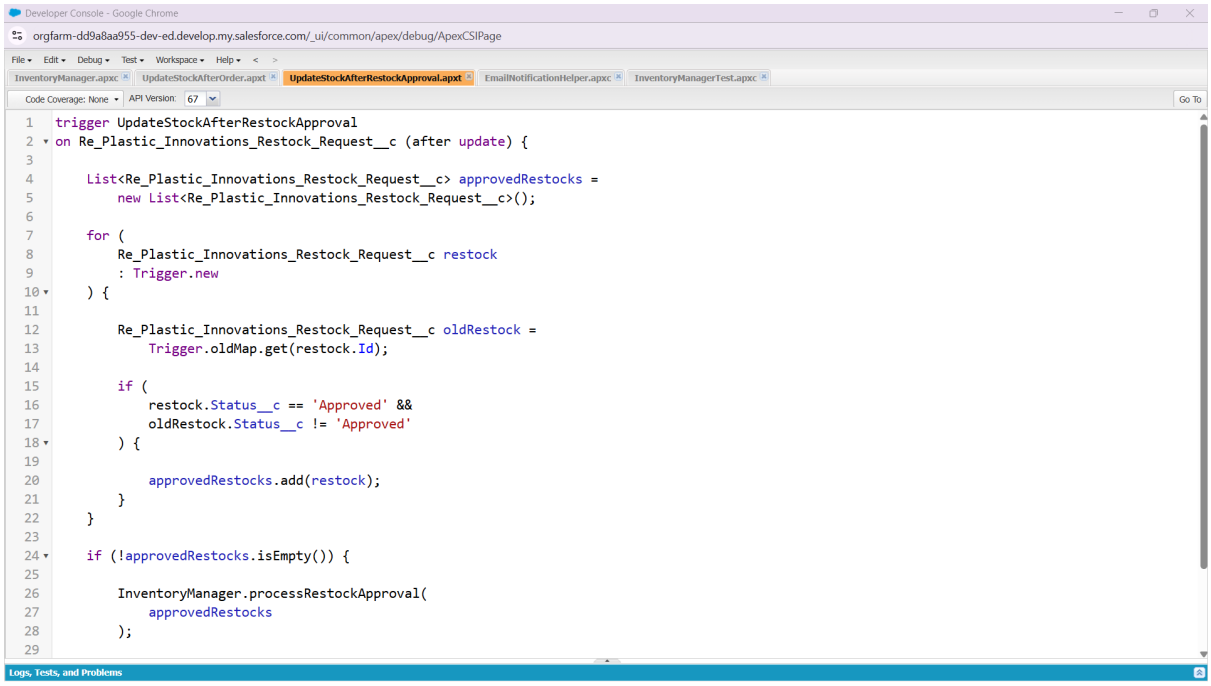


Figure 23: UpdateStockAfterRestockApproval Apex trigger.

2. A customer order is created.
3. Apex checks the available stock against the order quantity.
4. If stock is insufficient, a Restock Request is created with status Pending.
5. The Restock Request is reviewed.
6. The status is changed to Approved.
7. The approval trigger detects the status change.
8. Apex increases the related product stock using the approved requested quantity.

Business Result

The Apex automation connects order processing with restocking. This reduces manual stock calculations and ensures that approved replenishment quantities are reflected in the related recycled-product record.

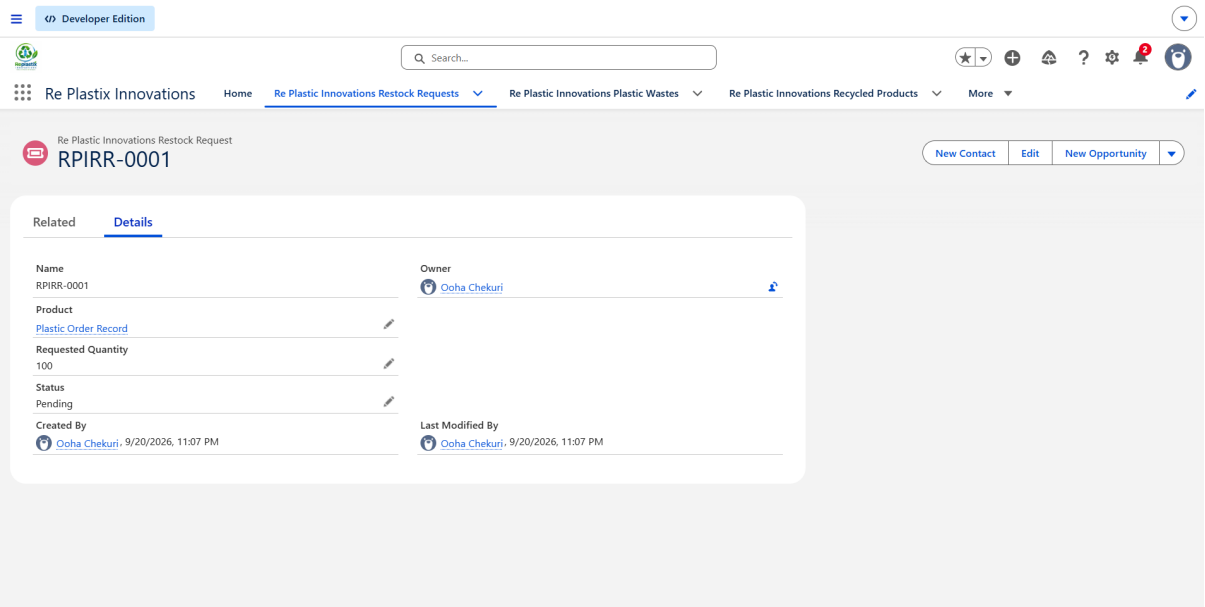


Figure 24: Restock Request created with Pending status.

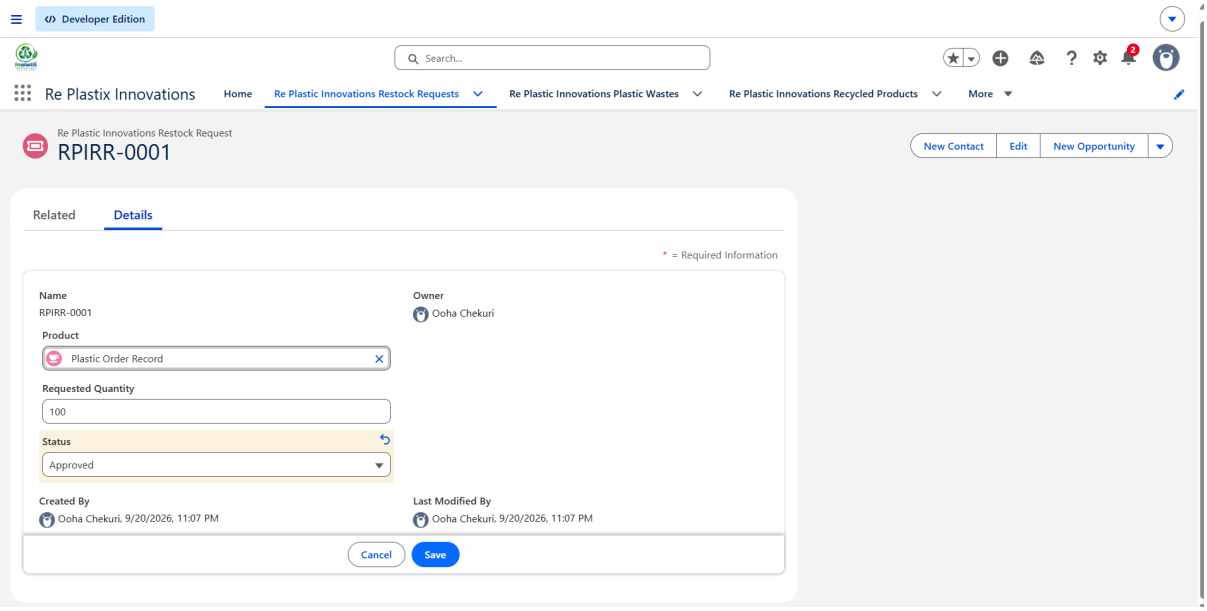


Figure 25: Restock Request after its status is changed to Approved.

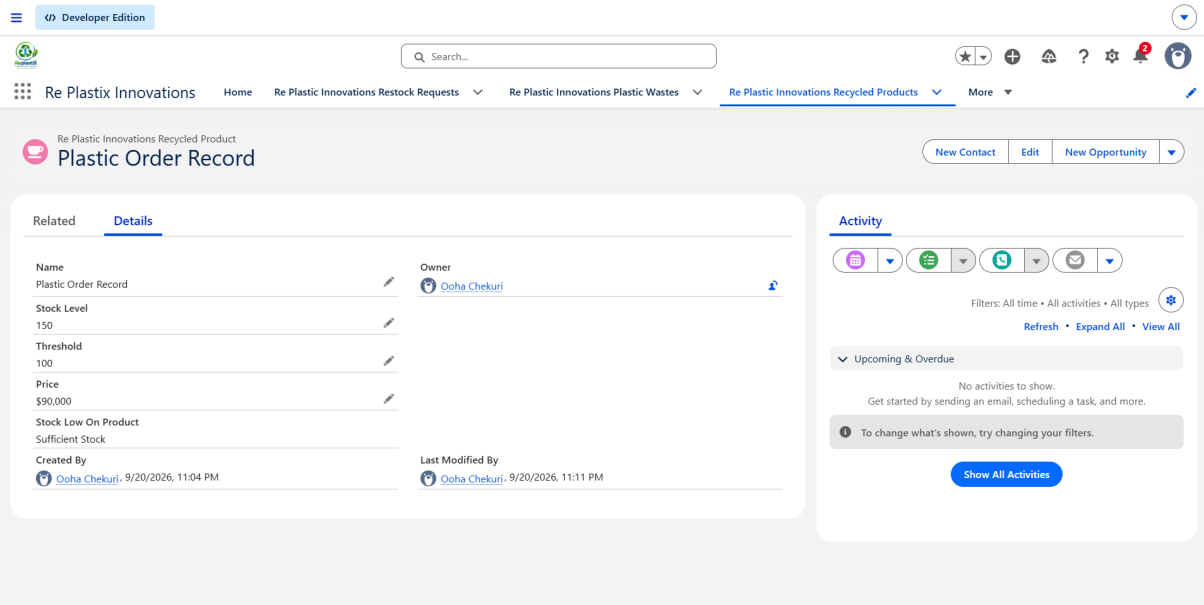


Figure 26: Recycled Product record showing updated stock after approval.

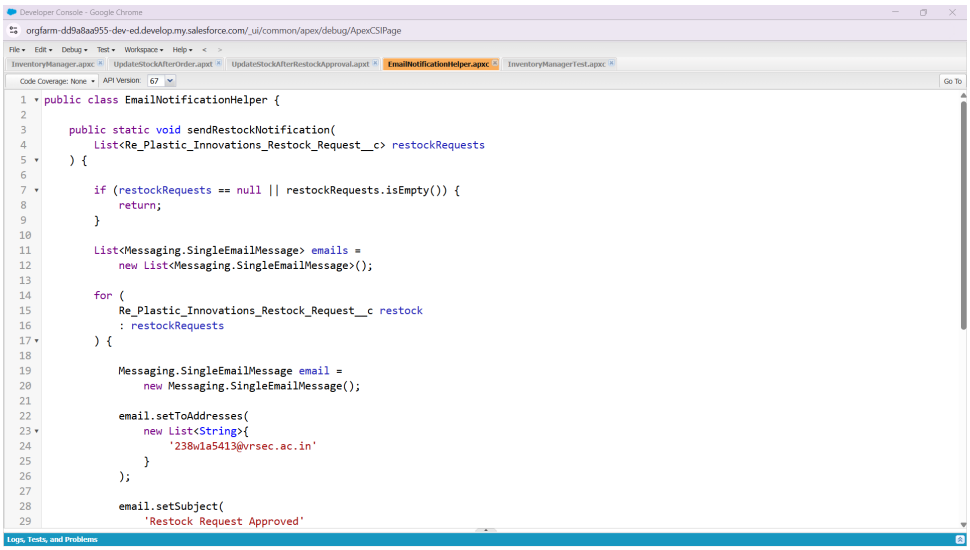
7.7 Email Notification Helper

An Apex helper class named `EmailNotificationHelper` was also created to send a notification after a restock request is approved.

The helper prepares an email with the subject:

Restock Request Approved

The message informs the recipient that the restock request has been approved and that the stock update can proceed.



```apex
public class EmailNotificationHelper {

    public static void sendRestockNotification(
        List<Re_Plastic_Innovations_Restock_Request__c> restockRequests
    ) {

        if (restockRequests == null || restockRequests.isEmpty()) {
            return;
        }

        List<Messaging.SingleEmailMessage> emails =
            new List<Messaging.SingleEmailMessage>();

        for (
            Re_Plastic_Innovations_Restock_Request__c restock
            : restockRequests
        ) {

            Messaging.SingleEmailMessage email =
                new Messaging.SingleEmailMessage();

            email.setToAddresses(
                new List<String>{
                    '238w1a5413@vrsec.ac.in'
                }
            );

            email.setSubject(
                'Restock Request Approved'
```

Figure 27: EmailNotificationHelper Apex class.

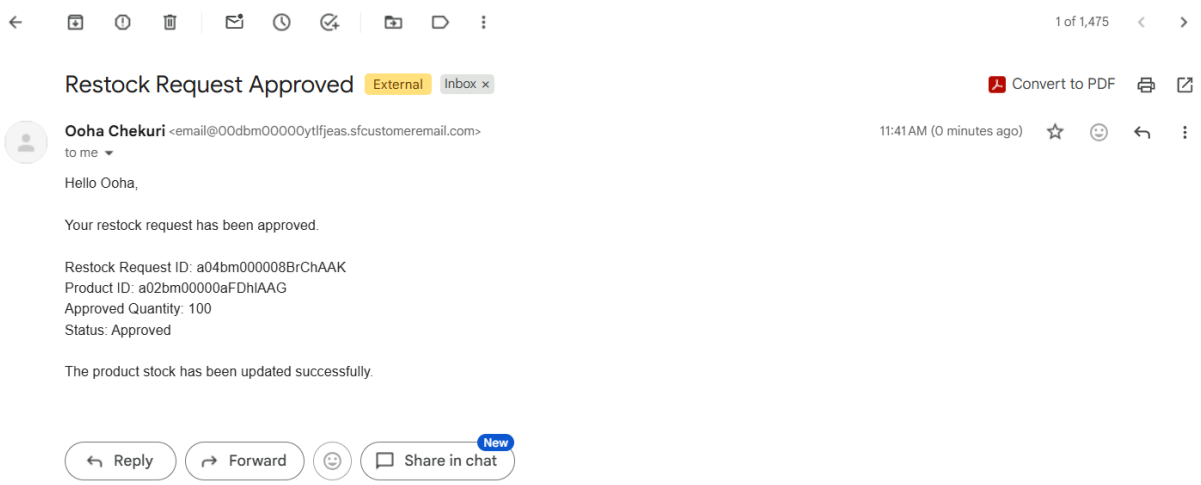


Figure 28: Email notification related to the approved restock request.

7.8 Apex Test Class

The Apex test class is named:

`InventoryManagerTest`

The test class creates test data for a recycled product and orders, then tests the inventory-processing and restock-approval methods.

The documented test activities are:

1. Create a recycled product with initial stock and threshold values.
2. Create orders with quantities representing sufficient and insufficient stock conditions.
3. Run the inventory-processing method.
4. Verify the resulting product stock.
5. Verify the creation of Pending Restock Request records.
6. Approve restock requests in the test data.
7. Run the restock-approval method.
8. Verify that the product stock increases by the approved requested quantity.
9. Run the Apex test and verify the test result and coverage.

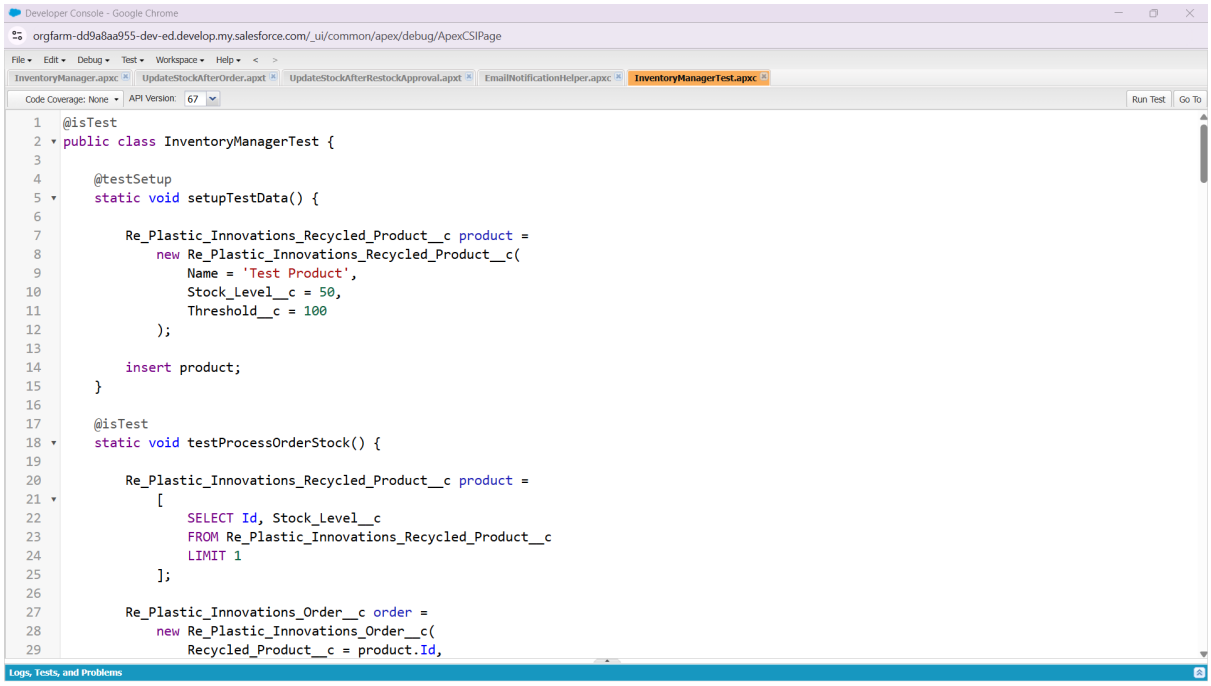


Figure 29: InventoryManagerTest Apex test class

8 Feature Testing Summary

The implementation was tested according to the behavior of the configured features.

Feature	Test Input / Action	Expected Result
Order Quantity Validation	Enter quantity as 0 or a negative value.	Salesforce displays the configured validation error and prevents the invalid record from being saved.
Collection Date Validation	Enter a collection date later than today's date.	Salesforce displays the configured validation error.
Low Stock Formula	Set stock below the threshold.	Formula displays "Low Stock - Restock Needed".
Low Stock Flow	Run the scheduled flow with stock below threshold.	A high-priority Task is created for the product owner.
Order Stock Processing	Create an order with sufficient stock.	Product stock is reduced by the order quantity.

Insufficient Stock	Create an order requiring more stock than is available.	A Pending Restock Request is created for the product.
Restock Approval	Change Restock Request status to Approved.	Related product stock is increased by the requested quantity.
Apex Tests	Run `InventoryManagerTest`.	Test methods execute and the configured test results/coverage are verified.

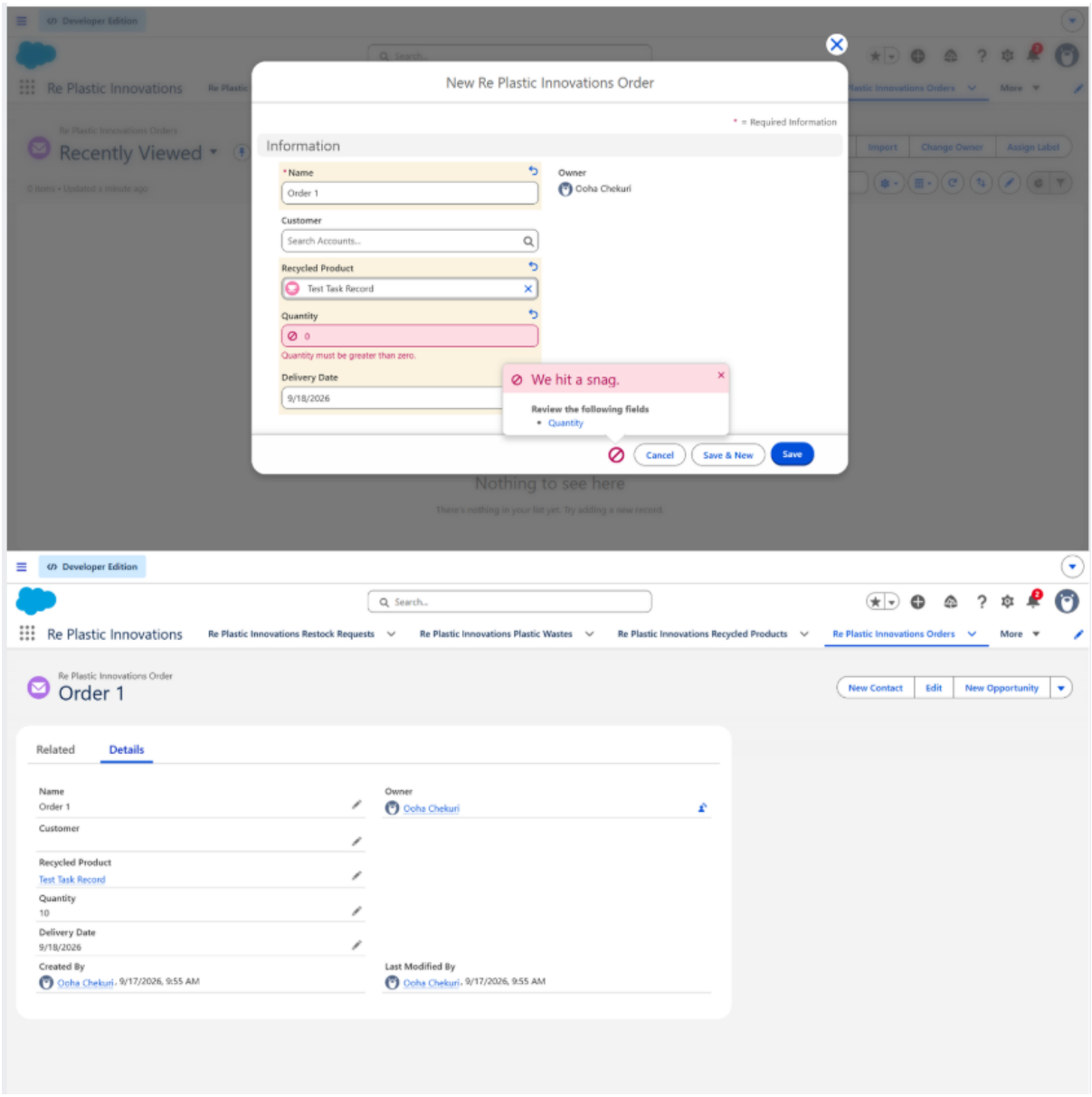


Figure 30: Validation-rule test results for zero-quantity

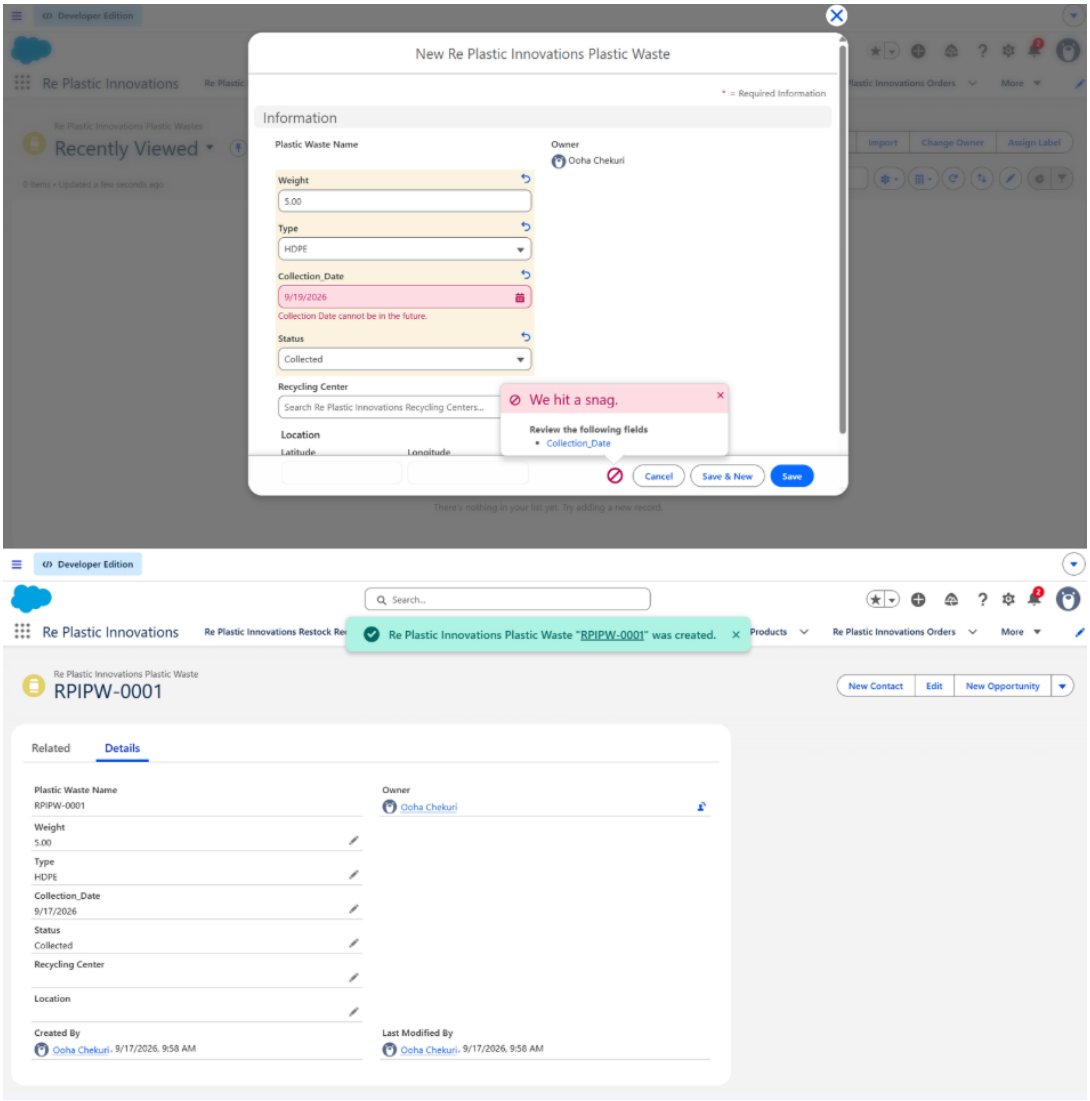


Figure 31: Validation-rule test results for future date

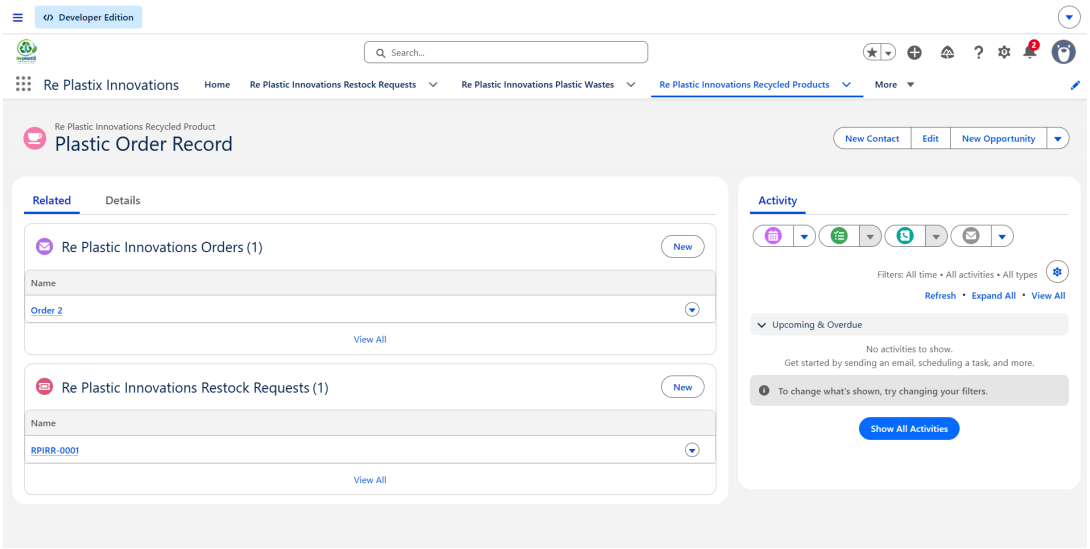


Figure 32: Automation test results for low-stock processing.

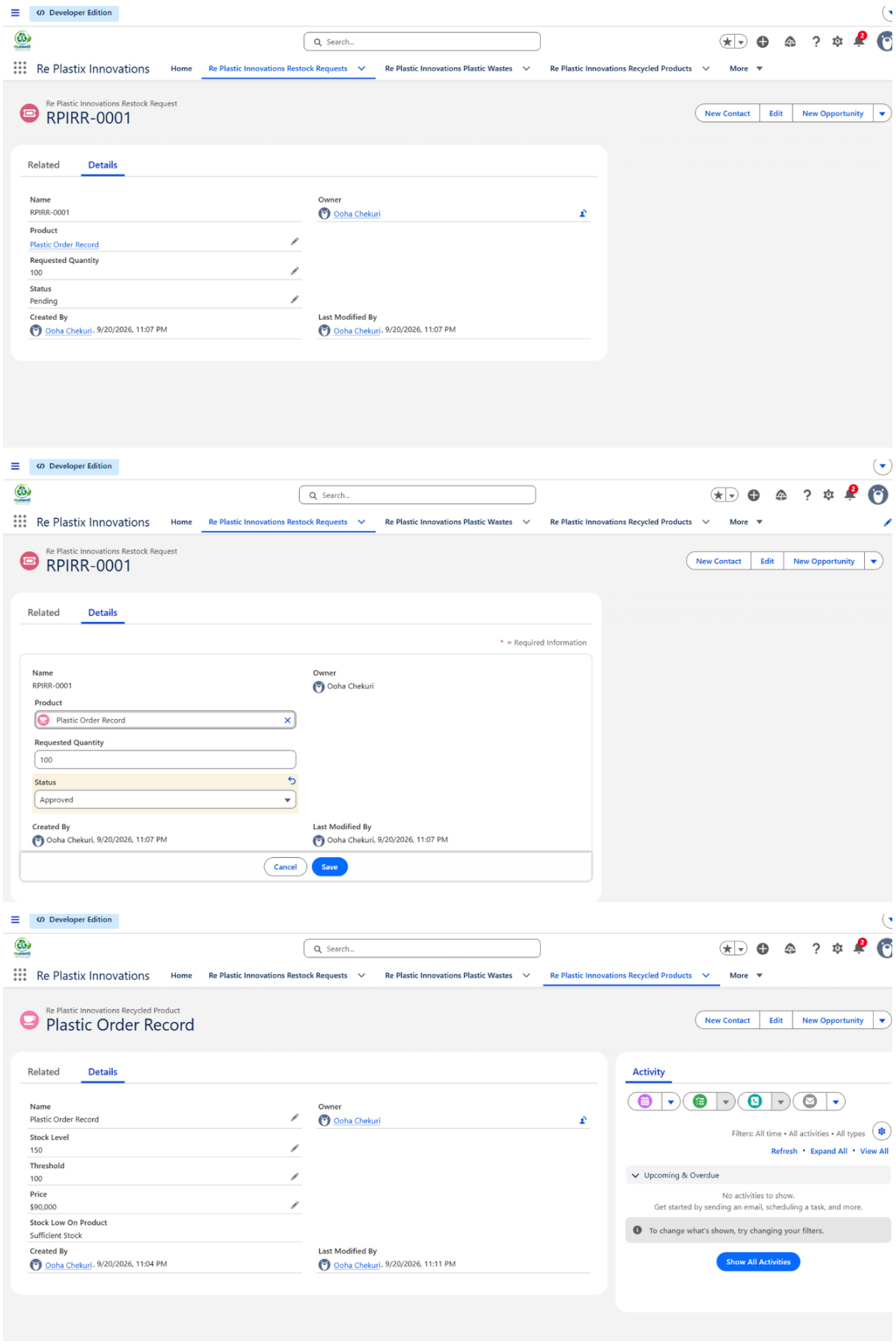


Figure 33: End-to-end Apex test showing order, restock request, approval, and stock update.

9 Conclusion

The Replastix Innovations Salesforce project provides a structured solution for managing plastic waste, recycling centers, recycled products, orders, and restock requests.

The project demonstrates Salesforce data modelling through custom objects, fields, and relationships. Data security is implemented using roles, profiles, users, and record-level access. Flow Builder provides automation for identifying low-stock products and creating tasks, while validation rules help maintain correct business data.

Apex extends the automation by processing orders, generating restock requests when inventory is insufficient, and updating product stock when a restock request is approved. The Apex test class provides a mechanism for verifying the inventory-processing logic.

Overall, the implemented Salesforce components provide an integrated data model, controlled access, validation, Flow Builder automation, and Apex-based inventory processing for the Replastix Innovations use case.